Practical 9

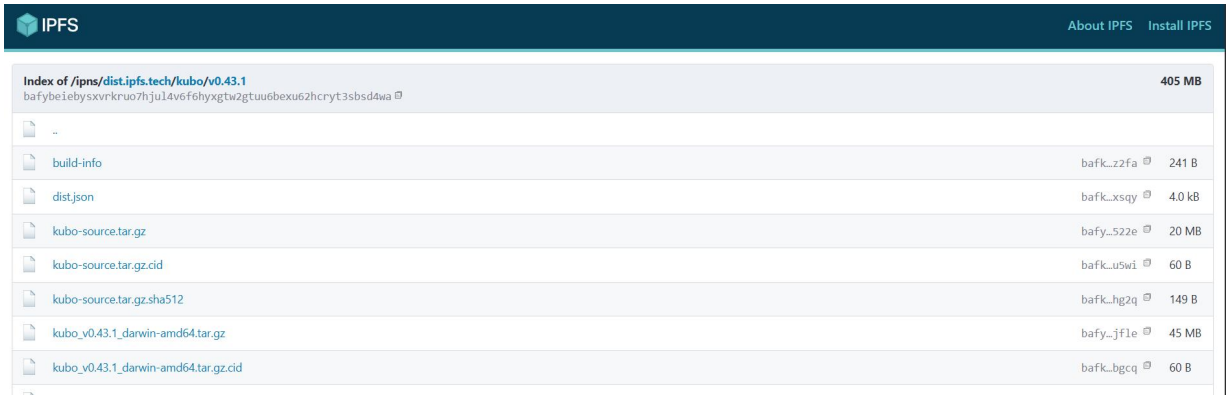
Aim : Install IPFS locally on our machine, initialize your node, view the nodes in network and add files and directories install Swarm and run any test file.

Introduction :

IPFS (InterPlanetary File System) is a decentralized, peer-to-peer file storage and sharing system. Instead of storing files on a single central server, IPFS stores and retrieves files using a unique **CID (Content Identifier)**. It allows users to share files directly between connected nodes.

Steps :

Step 1 : Download and install IPFS Kubo and check version.




```
C:\Users\chand>C:\kubo\ipfs.exe --version
ipfs version 0.43.1
```

Step 2 : Initialize Node.

```
C:\kubo>ipfs init
generating ED25519 keypair...done
peer identity: 12D3KooWAYLukArqfjiE2yAZaptjinAL49AhkAzLiH1VQatQKHkY
initializing IPFS node at C:\Users\chand\.ipfs
Error: ipfs configuration file already exists!
Reinitializing would overwrite your keys
```

Step 3 : Start IPFS.

```
C:\kubo>ipfs daemon
Initializing daemon...
Kubo version: 0.43.1-acc53c093
Repo version: 18
System version: amd64/windows
Golang version: go1.26.5
PeerID: 12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K
Swarm listening on 10.10.11.12:4001 (TCP+UDP)
Swarm listening on 10.40.104.133:4001 (TCP+UDP)
Swarm listening on 127.0.0.1:4001 (TCP+UDP)
Swarm listening on 169.254.140.15:4001 (TCP+UDP)
Swarm listening on 169.254.140.236:4001 (TCP+UDP)
Swarm listening on 169.254.151.7:4001 (TCP+UDP)
Swarm listening on [2409:40c1:4008:6672:1490:bd7a:27d4:1fe7]:4001 (TCP+UDP)
Swarm listening on [2409:40c1:4008:6672:50a8:13b1:481c:8753]:4001 (TCP+UDP)
Swarm listening on [::1]:4001 (TCP+UDP)
Run 'ipfs id' to inspect announced and discovered multiaddrs of this node.
RPC API server listening on /ip4/127.0.0.1/tcp/5001
WebUI: http://127.0.0.1:5001/webui
Gateway server listening on /ip4/127.0.0.1/tcp/8080
Routing V1 API exposed at http://127.0.0.1:8080/routing/v1
Daemon is ready
```

Step 4 : View Node and Network.

```
C:\kubo>ipfs id
{
        "ID": "12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
        "PublicKey": "CAESIJzUKE0x7kZmg9nW7Bp8DudTjfsLhaiJdxLpFcG+E9ho",
        "Addresses": [
                "/ip4/10.10.11.12/tcp/4001/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/10.10.11.12/udp/4001/quic-v1/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/10.10.11.12/udp/4001/quic-v1/webtransport/certhash/uEiDpUhLG2UPbFiCHCamxVrXjJtPQ5qV1Z1xBUWoJWdmxBw/certhash/uEiCj_yzpsxYzwG7TwF9CKpzTxFK6yqoy-a_Y18AXzv6-JQ/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/10.10.11.12/udp/4001/webrtc-direct/certhash/uEiCdl-9yRxm1nHzLUX9LVOt46B0BcevBk09t7A2SZGIVQg/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/10.40.104.133/tcp/4001/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/10.40.104.133/udp/4001/quic-v1/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/10.40.104.133/udp/4001/quic-v1/webtransport/certhash/uEiDpUhLG2UPbFiCHCamxVrXjJtPQ5qV1Z1xBUWoJWdmxBw/certhash/uEiCj_yzpsxYzwG7TwF9CKpzTxFK6yqoy-a_Y18AXzv6-JQ/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/10.40.104.133/udp/4001/webrtc-direct/certhash/uEiCdl-9yRxm1nHzLUX9LVOt46B0BcevBk09t7A2SZGIVQg/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/127.0.0.1/tcp/4001/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/127.0.0.1/udp/4001/quic-v1/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/127.0.0.1/udp/4001/quic-v1/webtransport/certhash/uEiDpUhLG2UPbFiCHCamxVrXjJtPQ5qV1Z1xBUWoJWdmxBw/certhash/uEiCj_yzpsxYzwG7TwF9CKpzTxFK6yqoy-a_Y18AXzv6-JQ/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/127.0.0.1/udp/4001/webrtc-direct/certhash/uEiCdl-9yRxm1nHzLUX9LVOt46B0BcevBk09t7A2SZGIVQg/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/169.254.140.15/tcp/4001/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/169.254.140.15/udp/4001/quic-v1/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/169.254.140.15/udp/4001/quic-v1/webtransport/certhash/uEiDpUhLG2UPbFiCHCamxVrXjJtPQ5qV1Z1xBUWoJWdmxBw/certhash/uEiCj_yzpsxYzwG7TwF9CKpzTxFK6yqoy-a_Y18AXzv6-JQ/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/169.254.140.15/udp/4001/webrtc-direct/certhash/uEiCdl-9yRxm1nHzLUX9LVOt46B0BcevBk09t7A2SZGIVQg/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/169.254.140.236/tcp/4001/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/169.254.140.236/udp/4001/quic-v1/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/169.254.140.236/udp/4001/quic-v1/webtransport/certhash/uEiDpUhLG2UPbFiCHCamxVrXjJtPQ5qV1Z1xBUWoJWdmxBw/certhash/uEiCj_yzpsxYzwG7TwF9CKpzTxFK6yqoy-a_Y18AXzv6-JQ/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
                "/ip4/169.254.140.236/udp/4001/webrtc-direct/certhash/uEiCdl-9yRxm1nHzLUX9LVOt46B0BcevBk09t7A2SZGIVQg/p2p/12D3KooWLNZQKA7B7YaK8Z5yGXbvpjAiftedrbUudtGNVbWX422K",
```

```
C:\kubo>ipfs swarm peers
/ip4/157.180.100.178/udp/4001/quic-v1/p2p/12D3KooWP8NiKbn5GZb9T1vZZqmgu9hA6hoyeMNQbQcEpp6yEbBp
/ip4/178.63.88.58/udp/4001/quic-v1/p2p/12D3KooWCDKDLEopE6G3tmLE7wpuL9nkijpXvwzuvV5K3W22iDDQ
/ip4/211.75.3.112/udp/4002/quic-v1/p2p/12D3KooWC2NBXHiCQ2n6Wm3ycPG29boziR7e8H2R4rrpHVX9RyaY
/ip4/46.255.204.203/udp/4001/quic-v1/p2p/12D3KooWSqMVaShoqJzP9q3cFe1tqafgxYpTryjPKzKzzLfYdF3M
/ip4/52.87.98.195/udp/9020/quic-v1/p2p/12D3KooWChddCemHfw4dCcay4TVCtmGFCVBq9rZqEd1J2HcThv3o
/ip4/65.108.15.30/udp/4002/quic-v1/p2p/12D3KooWHB1p5h283joX5Nu8WLczr3SyvDSPY4T6WssAGEfmYTqT
/ip4/65.109.109.140/udp/4001/quic-v1/p2p/12D3KooWBVPHMfEMb1Ak7osgFteuKgF8nSHWByAvqMFw5kS8pE8T
/ip4/80.145.114.79/tcp/4001/p2p/12D3KooWDf2yBQ5wSpkGWGQnQaNjs6ddm49KdWT8aBvAXC1UM8jR
/ip4/94.23.149.119/udp/4001/quic-v1/p2p/12D3KooWFLpeNew7KyAcwp9ZyfHCj9P7t8RnB6gbKYGECrpw7pT2
/ip6/2604:2dc0:200:484::1/udp/4001/quic-v1/p2p/QmNnooDu7bfjPFoTZYxMNLWUQJyrVwtbZg5gBMjTezGAJN
/ip6/2607:fdc0:81:30:5e6f:69ff:fe54:7d90/udp/59746/quic-v1/p2p/12D3KooWEuHxe4NtinP2bopLPd41vRBJaTNFy131csJCMFWoFciu
/ip6/2a07:e01:2::123/tcp/4001/p2p/12D3KooWDdCu4KezmnKGssgSk35eVjx1Md1fbqQVPY1XUn9twK6F
/ip6/64:ff9b::808c:4e6f/tcp/4001/p2p/12D3KooWKCKFAy9dxU8b2MV9CRPXuUEK2X3MyAxx76frqM2bSqFi
/ip6/64:ff9b::88f3:2efe/tcp/4001/p2p/12D3KooWNLyYE5L4DxDxF8s9523w24k8q1nrUmymYkM8SMqf6wQj
/ip6/64:ff9b::88f3:2efe/tcp/4002/p2p/12D3KooWPoyn3uBUY4ywZSLn8RG6fAthYhh6N53qjYUxDZuwFwhb
/ip6/64:ff9b::88f3:2efe/tcp/4003/p2p/12D3KooWK1pnazya1APPnyMLwPJ6GK2vcnA7pYgCjyRvQLUhmYXv
/ip6/64:ff9b::d057:8693/tcp/4001/p2p/12D3KooWMVu4aW6Qsku3twj2WxSdyQT3ogifRAWXwe98YwuBqmJr
```

Step 5 : Create and Add a Test File.

```
C:\kubo>echo Hello IPFS > test.txt

C:\kubo>ipfs add test.txt
added QmZJTQuzmo5WrQcTQAWkbHsrbKY9kpZ3yUvVDM3Uowr4v8 test.txt
13 B / 13 B [----------------------------------------------------------------------------------------------------------------------------------] 709 B/s 100.00% 0s
```

Step 6 : Read the File.

```
C:\kubo>ipfs cat  QmZJTQuzmo5WrQcTQAWkbHsrbKY9kpZ3yUvVDM3Uowr4v8
Hello IPFS
```

Step 7 : Add a Directory.

```
C:\kubo>mkdir myfolder

C:\kubo>echo File1 > myfolder\file1.txt

C:\kubo>ipfs add -r myfolder
added QmPKH2NNLqbvdi3ehN2BTKUp49QEPA34DbmyKqY3ZQq8D4 myfolder/file1.txt
added QmZXgsoXD1c8MhQP3sV9KpsKX7N8eMTF5f2ABC588zfcJp myfolder
8 B / 8 B [----------------------------------------------------------------------------------------------------------------------------------] 585 B/s 100.00% 0s
```

Step 8 : Test Swarm.

```
C:\kubo>ipfs swarm peers
/ip4/102.129.165.154/udp/4001/quic-v1/p2p/12D3KooWGgo9m9gVGpX8F9rZq7zTEWnmvS3jxup5677GA4dmFK6T
/ip4/109.176.207.171/udp/4001/quic-v1/p2p/12D3KooWQc4KkjX76rXh92o4NqpW5BG3xRJUottw1zD6d3C9SgEW
/ip4/112.164.216.74/tcp/40641/p2p/QmP4DwXcQHA8E8xFbb7vSeWqwzPMpTEGjLoAPMqhZqM97P
/ip4/116.202.49.39/udp/4001/quic-v1/p2p/12D3KooWCXoHgwmgCK4MsAYdsyGNCWmxWj271ePH28SnVV7ytcJC
/ip4/125.118.129.212/udp/33141/quic-v1/p2p/12D3KooWCCk5wxQa1V78dugRr7C1MyHGRFesBHVzjsbbf7hVLmiL
/ip4/129.153.77.57/udp/4001/quic-v1/p2p/12D3KooWNccohfuQxkTiQyTzZAJFS4R4geHq3n65K3V7NTVtYc16
/ip4/130.61.229.245/tcp/4001/p2p/12D3KooWSG6Pdtb5qJasASyjRUkKKPK7uNuoQHhi8TbV7k8sYqNR
/ip4/133.125.51.192/tcp/4001/p2p/12D3KooWSY9oCzDJnH7R21uUNofVpmNKFPCpmWbS1AtmDQVtiJY6
/ip4/135.181.17.165/udp/4001/quic-v1/p2p/12D3KooWE5ecb35hbUZ5oXL7w7n629dSgMGB15HnWv29aBi1a6CF
/ip4/136.52.122.110/tcp/4001/p2p/12D3KooWD2VnrHMWSD1BqmU3LNr1i2eT7gkpWwHwUd7vLJRMaUMp
/ip4/136.54.135.7/tcp/4001/p2p/12D3KooWQdRaSf9qk5hrCNmZcZx9JRxQBqKbXxvsMDmpPbsmsvjd
/ip4/139.180.202.187/tcp/4001/p2p/12D3KooWEHYa3bLE6a7b4NiAnx2buffYjtdN4Sqhjq5pDUbWzXKA
/ip4/142.132.252.29/tcp/4001/p2p/12D3KooWLT5zFZKuGCoF57i6CJbGmUF4LnX5HJfkT8UCjdKh3MqP
/ip4/144.126.155.159/tcp/4001/p2p/12D3KooWF6UqWjsJGapaKNVAnbHBsoXuBU6P2tKNHMzcaoZgohDr
/ip4/144.6.50.248/tcp/4001/p2p/12D3KooWGr1MQGnRNkcDnvefnra27m9YjzraRByqM1acqhknE9yH
/ip4/148.135.195.205/udp/4001/quic-v1/p2p/12D3KooWQ8mMcPXvjPJNbBcbE8iwDx23ibbv5ZeCSp6YHFbL28wM
/ip4/149.28.82.235/udp/4001/quic-v1/p2p/12D3KooWC8Z6TqJhw34Qg28DWcX3SS3rb7RpLPPV9Sbrxx9gL6Bt
/ip4/152.42.213.18/udp/4001/quic-v1/p2p/12D3KooWRNNhQ1HgCctDWDHkjvWEiWJoSz7LTBa5fpgYvbNxpUp4
/ip4/154.12.251.44/tcp/4001/p2p/12D3KooWHYBHVQ64PyUEWa8wghWfgnzVu96rx9VJhLeHFJWfj6Qn
/ip4/154.18.187.200/udp/4001/quic-v1/p2p/12D3KooWEd2CwPLf4YYnjeBe2yLpgQdeoBfKjGTVGFYhFgLjfDqi
/ip4/155.138.142.61/udp/4001/quic-v1/p2p/12D3KooWLRPeFFWG3Ay3JR11sQRPfKMXL1yyktUQXKNc8bRDcVJg
/ip4/157.180.100.178/udp/4001/quic-v1/p2p/12D3KooWP8NiKbn5GZb9T1vZZqmgu9hA6hoyeMNQbQcEpp6yEbBp
/ip4/157.180.125.143/udp/4001/quic-v1/p2p/12D3KooWMhmeVNrgPnd6wEm5QBanWQ9LWmdSgpSVTpi95a8Xbb4Z
/ip4/158.220.112.235/tcp/4001/p2p/12D3KooWGnBSdNTsfNnaSYXUyb3hR7wXWi9QdeXosPhSKXsx1jcN
/ip4/158.247.200.242/tcp/4001/p2p/12D3KooWAM1jDFK6wefibtQLYc5PPKYLEiib6VWJrKJNCpaxpWLx
/ip4/158.247.227.49/udp/4001/quic-v1/p2p/12D3KooWS4peYaKfAtJ6nByR3eQuhiesX2RcMjuY5owevaXi4fyu
/ip4/159.100.6.158/udp/4001/quic-v1/p2p/12D3KooWKxV3xUR5iyZadZNuHfvkjW6iEP8tghiiWySa8oc5Rpep
/ip4/159.100.6.220/udp/4001/quic-v1/p2p/12D3KooWGj6GGauXwrZVoXiYryeAEcRW7pAW28oHNDUunYMry6hY
/ip4/159.89.3.46/udp/4001/quic-v1/p2p/12D3KooWRpS9JGmhHh1VETdXgLfoS2gMDmFZNELvQBQam8XhjsQ8
/ip4/161.97.154.181/tcp/9020/p2p/QmXRVwQe4z5fj2vB4YQF6DR6cmP5bh3hciXfo3qYtpdvPD
/ip4/161.97.186.251/tcp/4001/p2p/12D3KooWRyrcKsckwGS7Yo4kKujZCqxN3ixzTKrnWFVib7at4jF1
/ip4/161.97.87.235/tcp/4001/p2p/12D3KooWPjjekoywVWg4quqKPPbKj97ZWC2Xc3Z8syr1NwxFoVEr
/ip4/162.0.225.185/udp/4001/quic-v1/p2p/12D3KooWJgJaYR9NcYQzFfHLjytJZKjpkRKjPV6SgxjSbhxdyRP2
/ip4/162.250.191.106/udp/4001/quic-v1/p2p/12D3KooWEwDcVyKTPQziP1PxuWQB6mK93Hg4RxhnYMZ8PChLXVZJ
/ip4/165.22.107.59/udp/4001/quic-v1/p2p/12D3KooWMWKydzmVL8Hju5beo8iDGLseJzrfKnh6ABuaJB17DfBK
/ip4/169.255.240.77/tcp/4001/p2p/12D3KooWLhZpaNjWaZEzynbc8dihpxLmAB9qm9XkMDCRpWhmDd38
/ip4/169.58.159.153/udp/4001/quic-v1/p2p/12D3KooWEFyxCj4BfEwesnyesk7vUAmBL96eYSrEhe2jxyzw23XL
/ip4/172.105.245.252/tcp/4001/p2p/12D3KooWHnDw4B8QhcD3gPVfNR4tz92FUs9nXPZGtjp29yb9LBna
/ip4/172.233.58.199/udp/4001/quic-v1/p2p/12D3KooWCeX4ke9Zdg4h2nu1CbvzM23qBvX4YbVdZDQcABENkYHD
/ip4/172.245.55.193/tcp/4001/p2p/12D3KooWJQDoGtTpBJorNPy2Km5HQRmBqieEnurowTGqdsf6QSep
/ip4/173.48.167.10/tcp/4001/p2p/12D3KooWPjWKgEcA34NB1ZWZ785zonSKKKtuW1gVBkyzRCejAcpV
/ip4/178.63.88.58/udp/4001/quic-v1/p2p/12D3KooWCDKDLEopE6G3tmLE7wpuL9nkijpXvwzuvV5K3W22iDDQ
/ip4/181.214.100.127/udp/4001/quic-v1/p2p/12D3KooWCTZCiCsMhfsvA6bndfAL37S2YjGpwQLLuevdxMzJoBP3
/ip4/181.215.18.103/udp/4001/quic-v1/p2p/12D3KooWAu5JX9wKYdfZJnrdXjnr82xLbNGn8cnzmu2DGxtuuazD
/ip4/183.156.160.221/tcp/4001/p2p/12D3KooWBcWS7KQ8GkrN29KGspGw6PZHxeY5XXyvG1D7TKypiqJ3
/ip4/185.232.71.228/udp/4001/quic-v1/p2p/12D3KooWDokaCJ3eQKuA4sTpaz5YEpkqPJL19NiTU524QUnWrgwt
/ip4/187.62.8.47/udp/13597/quic-v1/p2p/12D3KooWBvCyh8Vezjcwjhe2etkvjxw7g8hx9SW4jP5c94yXkyX1
/ip4/188.163.83.246/tcp/65066/p2p/QmbVHF4oMeddvZ5xVjiDDscVvt7X65oHfiqPmHfSV8Zn1P
/ip4/188.63.183.133/udp/46992/quic-v1/p2p/12D3KooWNvNzaAu4C3XPNGfbRFn6tZNeHe68HP1ji9aBZaLaAx8T
/ip4/189.245.117.193/tcp/4001/p2p/12D3KooWHYqrkYErZEJEA7ZZi5MpqJrDWNsBrPrSkijXR2HxCLYV
/ip4/192.142.15.100/udp/4001/quic-v1/p2p/12D3KooWJqbEcaY5VNKpKuwypmuDPEgbaackvU3ZjhuZn3Vp8QDV
/ip4/192.195.80.4/tcp/4105/p2p/12D3KooWLMP9cdpVq7pNmneC4bc8NN1d2cUUTTW29s68AVX8KNpJ
```

```
/ip4/94.130.180.233/tcp/4001/p2p/12D3KooWEk2Cbbi6chjLgoqdcPQHJP8845sU6YrrZ8RYtYKiXyAZ
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